

Maqsudur Rahman

Computing Ph.D. in Computer Science Candidate, Boise State University

Trustworthy AI / Secure Machine Learning / Quantum Machine Learning / Robust Medical and Biomedical AI

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Academic Profile

Computer Science educator and researcher with more than seven years of university-level teaching experience and an active research record in trustworthy artificial intelligence, secure and adversarial machine learning, quantum machine learning, robust medical image analysis, and hybrid quantum-classical learning systems. Currently pursuing a Computing Ph.D. in Computer Science at Boise State University, with expected completion in May 2027, and serving as a Graduate Research Assistant in the Department of Computer Science. My doctoral research develops reliable AI methods for safety-critical biomedical and medical-imaging applications under realistic data imperfections and security threats, including noisy labels, corrupted inputs, weak supervision, adversarial vulnerability, backdoor poisoning, and quantum trainability challenges. My long-term academic goal is to build trustworthy, interpretable, data-efficient, and secure AI systems for healthcare, biomedical computing, and other high-impact computing domains.

Research Interests

- **Trustworthy, robust, and secure AI:** noisy-label learning, uncertainty-aware modeling, adversarial robustness, backdoor poisoning defense, safety-critical model evaluation, and reliable deployment.
- **Quantum machine learning and hybrid quantum-classical systems:** variational quantum circuits, compact quantum representations, quantum neural networks, quantum trainability, and quantum-enhanced medical imaging.
- **Medical and biomedical AI:** robust medical image classification, biomedical decision support, multimodal and longitudinal clinical analysis, neurodegenerative disease analysis, and interpretable clinical representation learning.
- **AI systems and reproducibility:** reproducible benchmarking, systems-level support for hybrid quantum-classical learning, high-performance AI workflows, and practical evaluation pipelines.
- **Applied machine learning and data science:** natural language processing, cybersecurity analytics, social media analytics, information security, and data-driven computing applications.

Education

Boise State University , Boise, Idaho, USA	Aug 2024–Present
Computing Ph.D. in Computer Science Candidate	Expected completion: May 2027
Coursework CGPA: 3.89/4.00 ; Advisor: Dr. Jun Zhuang	
Research areas: Trustworthy AI, quantum neural networks, robust medical imaging, secure/adversarial AI.	
Comilla University , Bangladesh	2015–2017
M.Sc. (Engineering) in Information and Communication Technology	CGPA: 3.96/4.00
Comilla University , Bangladesh	2009–2015
B.Sc. (Engineering) in Information and Communication Technology	CGPA: 3.87/4.00

Academic Appointments

Graduate Research Assistant , Department of Computer Science, Boise State University, USA	Aug 2024–Present
Assistant Professor , Department of ICT, Comilla University, Bangladesh	Nov 2023–Jul 2024
Lecturer , Department of ICT, Comilla University, Bangladesh	Nov 2021–Oct 2023
Senior Lecturer , Department of CSE, Port City International University, Bangladesh	Jan 2020–Oct 2021
Lecturer , Department of CSE, Port City International University, Bangladesh	Apr 2017–Dec 2019
Research Assistant , Department of ICT, Comilla University, Bangladesh	Jul 2014–Jun 2016

Selected Ph.D.-Stage Research Contributions

- **NQNN - Noise-aware quantum neural networks for medical image classification:** developed a robust hybrid quantum-classical learning framework for noisy-label medical image classification using Fourier attenuation, uncertainty-based sample reweighting, and adaptive pooling within variational quantum circuits. Accepted/published in MICCAI 2025.
- **QAPNet - Quantum-attentive patchwise learning under corrupted inputs:** developed a quantum-attentive patchwise model using patch-level quantum encoding, attention-based multiple-instance aggregation, contrastive alignment, and prototype regularization for noisy-input medical image analysis. Accepted for AAAI 2026.
- **HQADNet - Secure and prototype-supervised quantum representation learning:** developed a contrastive prototype-supervised quantum representation learning framework for robust medical image classification against training-time backdoor poisoning. Manuscript under review at NeurIPS 2026.
- **Hybrid quantum-classical systems and trainability:** published systematic and empirical studies on systems-level support for hybrid quantum-classical learning and barren plateau mitigation strategies in quantum neural networks on standard and medical image datasets.

Teaching, Mentoring, and Student Supervision

- More than seven years of university-level teaching experience in Computer Science, ICT, AI/ML, cybersecurity, software, and core computing areas before beginning doctoral study.
- Taught or contributed to courses including Structured Programming, Object-Oriented Programming, Data Structures, Algorithms, Operating Systems, Information Security, Artificial Intelligence, Machine Learning, and Deep Learning.
- Supervised **56 undergraduate projects** and **15 thesis students**: 40 projects and 8 thesis students at Port City International University; 16 projects and 7 thesis students at Comilla University.
- Mentored students in programming, AI/ML applications, software development, data analysis, cybersecurity, natural language processing, research design, technical writing, and scholarly presentation.
- Teaching approach: concept-first explanation, laboratory reinforcement, reproducible implementation, project-based learning, research-oriented mentoring, and industry/application-aware computing education.

Research Grants

- **Principal Investigator**, “Feature-Based Online Social Media Bullying Detection using Point-wise Mutual Information (PMI) and Machine Learning on Bangla,” University Grants Commission (UGC) via Research Cell, Comilla University. 2022–2023
- **Principal Investigator**, “Categorization of Bangla Cyberbullying Texts on Social Media using Adaptive Neuro-Fuzzy Inference System: Corpus Creation and Detection,” University Grants Commission (UGC) via Research Cell, Comilla University. 2021–2022

Manuscript Under Review

1. **M. Rahman** and J. Zhuang, “HQADNet: Contrastive Prototype-Supervised Quantum Representation Learning for Robust Medical Image Classification against Backdoor Poisoning”. *Manuscript under review, NeurIPS 2026*.

Selected Ph.D.-Stage Peer-Reviewed or Accepted Publications

1. **M. Rahman** and J. Zhuang, “QAPNet: A Quantum-Attentive Patchwise Network for Robust Medical Image Classification Under Noisy Inputs”. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026. Accepted for presentation.
2. **M. Rahman** and J. Zhuang, “NQNN: Noise-aware Quantum Neural Networks for Medical Image Classification”. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, LNCS, Springer, 2025. Accepted/published.
3. **M. Rahman**, R. Liu, A. Majumder, P. C. Paul, K. Mo, A. Begum, K. Sultana, N. Akter, L. Wei, Y. Zhang, and J. Zhuang, “Benchmarking Barren Plateau Mitigation Strategies in Quantum Neural Networks on Standard and Medical Image Datasets”. *Journal of Imaging*, vol. 12, Article 275, 2026.
4. **M. Rahman**, P. C. Paul, A. Begum, K. Sultana, N. Akter, A. Majumder, M. Zhu, Z. Sheng, W. Xin, X. Jin, and J. Zhuang, “Systems-Level Support for Hybrid Quantum-Classical Learning: A Systematic Review with a Medical Imaging Translation Lens”. *Journal of Imaging*, vol. 12, Article 232, 2026.

Earlier Peer-Reviewed Journal Articles

1. **M. Rahman**, M. T. Ahmed, S. Nur, and A. Z. M. Islam, “The Prediction of COVID-19 Pandemic Outbreak on Bangladesh Perspective Using Multiple Machine Learning Algorithms: A Comparative Study”. *International Journal of Electrical and Computer Engineering (IJECE)*, 2022.
2. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, “NLP and Machine Learning Based Cyberbullying Detection for Bangla and Romanized Bangla Texts”. *TELKOMNIKA (Telecommunication Computing Electronics and Control)*.
3. K. Farhana, **M. Rahman**, and M. T. Ahmed, “An Intrusion Detection System for Packet and Flow Based Networks using Deep Neural Network Approach”. *International Journal of Electrical and Computer Engineering (IJECE)*.
4. P. C. Paul, **M. Rahman**, A. Begum, M. T. Ahmed, D. Chakraborty, and M. S. Rahman, “Combining BERT with LDA: Improved Topic Modeling in Bengali Language”. *IAENG International Journal of Computer Science*, vol. 52, no. 2, pp. 383–393, 2025.
5. **M. Rahman** and M. Shahjahan, “Exploring the Taxonomy of Survey Papers on Large Language Models Using Classical Machine Learning”. *Journal of Intelligent Learning Systems and Applications*, vol. 17, no. 2, pp. 108–122, 2025.
6. A. H. Antar, M. T. Ahmed, M. Akter, M. S. Rahman, **M. Rahman**, P. C. Paul, and N. Parvin, “Deep Neural Network Based Spam Email Classification Using Attention Mechanisms”. *Journal of Intelligent Learning Systems and Applications*, vol. 15, pp. 144–164, 2023.
7. M. T. Ahmed, N. Akter, **M. Rahman**, D. Das, T. A. Z. M., and G. Rashed, “Multimodal Cyberbullying Meme Detection from Social Media Using Deep Learning Approach”. *International Journal of Computer Science and Information Technology (IJCSIT)*, vol. 15, pp. 27–37, 2023.
8. M. T. Ahmed, A. H. Antar, **M. Rahman**, A. Z. M. T. Islam, D. Das, and M. G. Rashed, “Social Media Cyberbullying Detection on Political Violence from Bangla Texts Using Machine Learning Algorithm”. *Journal of Intelligent Learning Systems and Applications*, vol. 15, no. 4, pp. 108–122, 2023.
9. M. T. Ahmed, A. A. Rubi, S. Rahman, and **M. Rahman**, “Red-AODV: A Prevention Model of Black Hole Attack for VANET Protocols and Identification of Malicious Nodes in VANET”. *International Journal of Computer Networks and Applications (IJCNA)*, 2021.

Peer-Reviewed Conference Papers

1. **M. Rahman**, S. Nur, M. T. Ahmed, D. Das, and A. Z. M. Islam, "A Feature Engineering Approach for Detecting Cyberbullying in Bangla Text Using Machine Learning". *International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*, 2022.
2. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, "Introduction of PMI-SO Integrated with Predictive and Lexicon Based Features to Detect Cyberbullying in Bangla Text Using Machine Learning". *International Conference on Artificial Intelligence: Advances and Applications (ICAIAA)*, 2021.
3. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, "Deployment of Machine Learning and Deep Learning Algorithms in Detecting Cyberbullying in Bangla and Romanized Bangla Text: A Comparative Study". *IEEE International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, 2021.
4. A. S. Urmi, M. T. Ahmed, **M. Rahman**, and A. Z. M. Islam, "A Proposal of Systematic SMS Spam Detection Model Using Supervised Machine Learning Classifiers". *International Conference on Computer Vision and Robotics (CVR)*, 2021.
5. S. Palit, S. Nur, Z. Khatun, **M. Rahman**, and M. T. Ahmed, "Analysis of Online Education System of Bangladesh during COVID-19 Pandemic Based on NLP and Machine Learning: Problem and Prospect". *International Conference on Emerging Trends in Industry 4.0 (ETI 4.0)*, 2021.
6. M. Islam, **M. Rahman**, M. T. Ahmed, A. Z. M. Islam, D. Das, and M. M. Hoque, "Sexual Harassment Detection using Machine Learning and Deep Learning Techniques for Bangla Text". *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2023.
7. P. C. Paul, M. S. Uddin, M. T. Ahmed, M. M. Hoque, and **M. Rahman**, "Semantic Topic Extraction from Bangla News Corpus Using LDA and BERT-LDA". *International Conference on Computer and Information Technology (ICCIT)*, 2022.
8. I. Hossain, **M. Rahman**, T. Ahmed, and A. Z. M. Islam, "Forecasting the Rating of Online Products from Customer Text Review based on Machine Learning Algorithms". *International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*.
9. I. Hossain, **M. Rahman**, T. Ahmed, A. Z. M. Islam, and S. Rahman, "Rating Prediction of Product Reviews of Bangla Language Using Machine Learning Algorithms". *International Conference on Artificial Intelligence and Mechatronics Systems (AIMS)*, 2021.
10. P. Barman, **M. Rahman**, and M. T. Ahmed, "A Comparison of Statistical and Deep Learning Model in Forecasting Time Series". *International Conference on Sustainable Development in Technology for 4th Industrial Revolution (ICSdTIR)*.
11. P. Shil, U. S. Rahman, **M. Rahman**, and M. S. Islam, "An Approach for Detecting Bangla Spam Comments on Facebook". *International Conference on Electronics, Communications and Information Technology (ICECIT)*, 2021.
12. M. Rahman, M. S. Rahman, N. Parvin, and **M. Rahman**, "Fundamental Frequency Extraction by Utilizing the Modified Weighted Autocorrelation Function in Noisy Speech". In: *International Conference on Advances in Data-driven Computing and Intelligent Systems*, Springer, 2023.

Book Chapter

1. **M. Rahman** and S. Hossain, "Performance Evaluation of Social-Aware Routing Protocols in an Opportunistic Network". In: *Opportunistic Networks: Mobility Models, Protocols, Security and Privacy*.

Honors and Awards

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| • Best Paper Award Finalist, ICAIAA 2022. | 2022 |
| • Best Paper Award Finalist, IEEE ICAECT 2021. | 2021 |

Professional Service and Leadership

- Reviewer and service contributor for peer-reviewed venues in artificial intelligence, machine learning, medical imaging, data science, cybersecurity, and computing.
- Moderator, Comilla University Science Club.
- Advisory Member, Comilla University IT Society.
- Advisor, ICT Association, Department of ICT, Comilla University.
- Coach, ACM ICPC student programming team, Comilla University.
- Participated in approximately 15 academic workshops and seminars on outcome-based education, curriculum design, research methodology, scholarly writing, artificial intelligence, machine learning, deep learning, cybersecurity, data mining, NLP, and software engineering.

Technical Skills

Programming and software: Python, C, C++, Java, Git/GitHub, Jupyter, Linux, Microsoft Office, Overleaf, L^AT_EX.

Machine learning and deep learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, SciPy, Matplotlib.

Quantum computing: PennyLane, Qiskit, variational quantum circuits, quantum neural networks, hybrid quantum-classical pipelines, cuQuantum.

Medical imaging and computer vision: OpenCV, MedMNIST, OASIS, image classification, biomedical image analysis, patch-based learning, robust evaluation.

Trustworthy and secure AI: noisy-label learning, uncertainty estimation, adversarial robustness, backdoor poisoning defense, prototype learning, robust representation learning.

References

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